Supplemental working source for "Prescribed finite-window reaction-time modality by conserved-budget support design"

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Internal status—not a submission artifact. This supplement consolidates the analytical statements that are otherwise distributed across the working manuscript and theorem notes. Section S2 derives the complete fixed-budget state and observable sensitivity hierarchy. Section S3 proves compact-positive-time convergence of the budget-normalized Doi reaction density to the exact free-exposure law through every prescribed finite time/control mixed jet. Section S4 retains an earlier, more general time-varying-variance construction that proves only a local lower bound of at least m modes. Section S5 gives the complete exact-m proof used by the theorem-first main working paper in the stationary-midpoint, common-variance subfamily: a global $2m-1$ zero bound, adjacent isolation, full posterior-sector complement certificate, slow-factor preservation, and fixed-ε positive-budget Doi transfer. For every fixed finite integer physical dimension $d \geq 2$ and prescribed fixed finite m, it yields exactly m nondegenerate maxima and $m-1$ nondegenerate minima on the declared compact window after taking the limits in the order ε first and positive reaction budget B second. All initial-law, variance, whole-window contact-interior, compact-weight, and functional-space hypotheses are explicit. The exact result is an existence theorem for an m-dependent support design, not same-support mode-count switching by allocation. No positive-budget finite-parameter topology, GIG-to-Doi realization, event-mass floor, or Lean verification of the analytic arguments is asserted here. Section S6 records the control-free 12-row source, free-axis, and packed-rate preparation, with explicit exclusions of killing, propagation, topology, and positive-budget execution.

S1. MODEL, BUDGET, AND ANALYTICAL SCOPE

We work with the Doi volume-reaction model [1] after the exact symmetry reduction appropriate to longitudinal catalyst slabs. Fix a finite integer physical dimension $d \geq 2$; the quotient has dimension $d+1$. Its bounded and unbounded versions are

$$\mathcal{Q}_d^{\mathrm{box}} = I_z \times I_{\parallel} \times \mathbb{T}_W^{d-1}, \tag{S1}$$
$$\mathcal{Q}_d^{\infty} = \mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}. \tag{S2}$$

The transverse quotient is exact in every fixed finite dimension. Haar invariance gives, for integrable h,

$$\int_{\mathbb{T}_W^{d-1}} \int_{\mathbb{T}_W^{d-1}} h(y_1 - y_2)\, \mathrm{d}y_1 \mathrm{d}y_2 = W^{d-1} \int_{\mathbb{T}_W^{d-1}} h(r_\perp)\, \mathrm{d}r_\perp. \tag{S3}$$

Thus no global transverse midpoint coordinate on the torus is introduced. The variables are $x = (z, r_\parallel, r_\perp)$, where z is the longitudinal midpoint and r is the relative coordinate. For two equal-diffusivity walkers, the free forward operator is

$$\mathcal{L}q = -\nabla \cdot (bq) + \nabla \cdot (\boldsymbol{D} \nabla q), \qquad \boldsymbol{D} = \mathrm{diag}(D/2, 2D, \ldots, 2D), \tag{S4}$$

with

$$b(x) = \big(-\gamma(z - \bar{z}), -\gamma r_\parallel, 0, \ldots, 0\big). \tag{S5}$$

On $\mathcal{Q}_d^{\mathrm{box}}$, the nonperiodic faces carry the zero-forward-flux condition

$$(bq - \boldsymbol{D} \nabla q) \cdot n = 0, \tag{S6}$$

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and the transverse coordinates are periodic. On $\mathcal{Q}_d^\infty$, natural decay is encoded through the weighted space specified below, not through an artificial reflecting box.

Let $0 < a < W/2$ and let $\chi_a(r)$ be the indicator of the embedded minimum-image d-ball $\{|r|_{\mathrm{mi}} < a\}$. Fixed nonnegative longitudinal profiles satisfy

$$\phi_j \in L^\infty, \qquad \int \phi_j(z)\,\mathrm{d}z = 1. \tag{S7}$$

The integral in Eq. (S7) is over I_z for $\mathcal{Q}_d^{\mathrm{box}}$ and over $\mathbb{R}$ for $\mathcal{Q}_d^\infty$. Define

$$V_j(z,r) = W^{-(d-1)}\chi_a(r)\phi_j(z), \qquad V_w = \sum_{j=1}^{J} w_j V_j, \tag{S8}$$

where $w_j \geq 0$ and $\sum_j w_j = 1$. The killing field is $K_{B,w} = BV_w$. Because the factor $W^{-(d-1)}$ normalizes the omitted transverse common-centre directions, the installed centre-space catalyst amount is exactly

$$\int_{\text{centre space}} \kappa_{B,w}(c)\,\mathrm{d}c = B. \tag{S9}$$

Equation (S9), rather than the integral of $K_{B,w}$ over all relative configurations, is the conserved resource. Supports, transport, contact radius, and the initial law remain fixed when w varies. If the local killing field has unit T^{-1}, then $[B] = L^d T^{-1}$ in physical dimension d. Thus the theorem's dimensional budgets cannot be compared across dimensions by assigning them the same numerical value without a separately declared nondimensionalization.

The initial density is independent of B, w, and every control coordinate. Throughout Secs. S2–S3 we assume

$$q_0 \geq 0, \qquad \langle 1, q_0\rangle = 1, \qquad q_0 \in \begin{cases} L^2(\mathcal{Q}_d^{\mathrm{box}}), & \text{on the bounded quotient,} \\ X_\pi, & \text{on the unbounded quotient,} \end{cases} \tag{S10}$$

where X_π and the normalized density π are defined below. The operator estimates remain valid for arbitrary signed or complex q_0 in the corresponding Hilbert space; nonnegativity and unit mass are imposed only for the survival and reaction-density interpretation.

With

$$A_{B,w} = \mathcal{L} - BM_{V_w}, \qquad q_{B,w}(t) = \mathrm{e}^{tA_{B,w}} q_0, \tag{S11}$$

the reaction-time density and survival probability obey

$$f_B(t,w) = B\langle V_w, q_{B,w}(t)\rangle, \tag{S12}$$

$$S_B(t,w) = \langle 1, q_{B,w}(t)\rangle, \qquad -\partial_t S_B = f_B. \tag{S13}$$

The differential identity in Eq. (S13) is classical for $t > 0$ and holds in integrated form from $t = 0$. The pairing is the integral of observable times density. Occupation-time and Feynman–Kac interpretations of this reactivity-times-occupancy identity are standard [2, 3]; the results below concern its conserved-budget mixed jets and a direct physical multimode construction.

The normalized reversible free density is

$$\pi(x) = Z_\pi^{-1} \exp\left[-\frac{\gamma(z-\bar{z})^2}{D} - \frac{\gamma r_\parallel^2}{4D}\right], \tag{S14}$$

uniform in the torus coordinates; its normalized transverse factor is $W^{-(d-1)}$. On $\mathcal{Q}_d^{\mathrm{box}}$, π is bounded above and away from zero and the state space is $X = L^2(\mathcal{Q}_d^{\mathrm{box}})$. On $\mathcal{Q}_d^\infty$, the density space and its observable dual are

$$X_\pi = L^2(\pi^{-1}\mathrm{d}x), \qquad X_\pi^* = L^2(\pi\mathrm{d}x). \tag{S15}$$

The map $u \mapsto q = \pi u$ is unitary from $L^2(\pi\mathrm{d}x)$ to X_π. The unbounded results require $q_0 \in X_\pi$; unweighted L^2 alone is not sufficient.

All time derivatives in this supplement are restricted to a compact window $[\tau, T] \subset (0,\infty)$. We do not claim a mixed-jet estimate at $t = 0$, on the long scale $t = O(B^{-1})$, or uniformly in the narrow-noise parameter used in Sec. S4.

S2. EXACT FIXED-BUDGET SENSITIVITY HIERARCHY

S2.1. Budget-tangent coordinates and first variation

For normalized patches, the tangent space of the conserved simplex is

$$\mathsf{T} = \{h \in \mathbb{R}^J : \mathbf{1}^\mathsf{T} h = 0\}. \tag{S16}$$

Fix a positive-definite control metric M and a matrix $P \in \mathbb{R}^{J\times(J-1)}$ satisfying

$$\mathbf{1}^\mathsf{T} P = 0, \qquad P^\mathsf{T} M P = I_{J-1}. \tag{S17}$$

Near a strictly interior control w_*, write

$$w(\theta) = w_* + P\theta, \qquad U_i = \sum_{j=1}^{J} P_{ji} V_j. \tag{S18}$$

The basis and nondimensionalization are frozen before reporting singular values. Changing coordinates without transforming the metric changes their numerical values.

For a fixed-support perturbation H, Duhamel's identity gives the exact first variation

$$\begin{aligned} D_V f_B(t)[H] &= B\langle H, \mathrm{e}^{tA_{B,w}} q_0\rangle \\ &\quad - B^2 \int_0^t \left\langle V_w, \mathrm{e}^{(t-s)A_{B,w}} M_H \mathrm{e}^{sA_{B,w}} q_0 \right\rangle \mathrm{d}s. \end{aligned} \tag{S19}$$

This is an amplitude derivative. It is not a shape derivative for a moving contact set or catalyst support.

S2.2. All finite state and observable sensitivities

For a multi-index $\beta \in \mathbb{N}_0^{J-1}$, let $q_\beta = \partial_\theta^\beta q_{B,w(\theta)}$, with $q_{\mathbf{0}} = q$. Write e_i for the ith coordinate multi-index.

Theorem S2.1 (Affine-control sensitivity hierarchy)**.** *For every finite multi-index β, the state sensitivity is the unique mild solution of*

$$\partial_t q_{\mathbf{0}} = A_{B,w(\theta)} q_{\mathbf{0}}, \qquad q_{\mathbf{0}}(0) = q_0, \tag{S20}$$

$$\partial_t q_\beta = A_{B,w(\theta)} q_\beta - B\sum_{i=1}^{J-1} \beta_i M_{U_i} q_{\beta-e_i}, \qquad q_\beta(0) = 0 \quad (|\beta| > 0). \tag{S21}$$

Terms with $\beta_i = 0$ vanish. For every integer $r \ge 0$ and $t > 0$, the exact observable mixed jet is

$$\partial_t^r \partial_\theta^\beta f_B = B\left[\langle V_{w(\theta)}, \partial_t^r q_\beta\rangle + \sum_{i=1}^{J-1} \beta_i \langle U_i, \partial_t^r q_{\beta-e_i}\rangle\right]. \tag{S22}$$

Proof. For bounded multiplication operators, the Dyson–Phillips series is norm-convergent on every compact time set and is entire in the finite affine control amplitudes. Hence it may be differentiated term by term. Since

$$\partial_{\theta_i} A_{B,w(\theta)} = -BM_{U_i}, \qquad \partial_{\theta_i}\partial_{\theta_j} A_{B,w(\theta)} = 0, \tag{S23}$$

the multi-index product rule applied to $\partial_t q = A_{B,w(\theta)} q$ produces exactly the multiplicity β_i in Eq. (S21). The initial law is control-independent, which gives the zero initial data for every positive order. Applying the same product rule to $f_B = B\langle V_{w(\theta)}, q\rangle$ yields Eq. (S22), because the affine observable has no control derivatives beyond first order. Analytic-semigroup smoothing, proved in Sec. S3, permits every fixed number of time derivatives for $t > 0$. □

At first and second order, Theorem S2.1 gives

$$\partial_t s_i = As_i - BU_i q, \qquad s_i(0) = 0, \tag{S24}$$

$$\partial_t s_{ij} = As_{ij} - BU_i s_j - BU_j s_i, \qquad s_{ij}(0) = 0, \tag{S25}$$

where multiplication symbols have been suppressed. Correspondingly,

$$\partial_{\theta_i} f_B = B\{\langle U_i, q\rangle + \langle V_w, s_i\rangle\}, \tag{S26}$$
$$\partial_{\theta_i\theta_j} f_B = B\{\langle U_i, s_j\rangle + \langle U_j, s_i\rangle + \langle V_w, s_{ij}\rangle\}. \tag{S27}$$

The first term in Eq. (S26) and the first two terms in Eq. (S27) are direct derivatives of the reaction observable. A sensitivity implementation that differentiates only the state omits these terms and computes the wrong control jet.

For any $r \geq 0$, Eqs. (S26)– (S27) remain valid after replacing every state by its rth time derivative. In particular, a cusp calculation has access to

$$\{f_t, f_{tt}, f_{ttt}, f_{tttt}, f_{t\theta_i}, f_{tt\theta_i}, f_{ttt\theta_i}\}, \tag{S28}$$

and second control derivatives remain available for continuation diagnostics. The safe time-derivative identity is

$$\partial_t^r f_B = B\langle V_w, A_{B,w}^r q(t)\rangle. \tag{S29}$$

No application of $(A_{B,w}^*)^r$ to the discontinuous contact indicator is required or claimed.

S2.3. Projected response under one conserved cost

For completeness, let Y_a be a finite collection of time/control observables and let $G_{aj} = \partial_{w_j} Y_a$. If a general patch basis has cost covector $c_j = \int \phi_j$, its budget-tangent response is

$$\widetilde{G} = G - \frac{(GM^{-1}c)c^{\mathsf{T}}}{c^{\mathsf{T}}M^{-1}c}. \tag{S30}$$

When $\widetilde{G}$ has full row rank, the unique minimum-M-norm budget-preserving infinitesimal control producing y is

$$h_* = M^{-1}\widetilde{G}^{\mathsf{T}}(\widetilde{G}M^{-1}\widetilde{G}^{\mathsf{T}})^{-1}y. \tag{S31}$$

This is constrained linear algebra only: it is local and supplies no global finite-amplitude controllability statement.

S3. COMPACT-POSITIVE-TIME WEAK-REACTION MIXED JETS

S3.1. Analytic killed semigroups

Proposition S3.1 (Bounded and unbounded analytic realizations)**.** *On $\mathcal{Q}_d^{\mathrm{box}}$, realize $\mathcal{L}$ with Eq.* (S6) *and periodic transverse conditions on $X = L^2(\mathcal{Q}_d^{\mathrm{box}})$. It generates a positive analytic semigroup. For every real $B \geq 0$ and real simplex control w, $A_{B,w}$ has the same domain and generates a positive, mass-decreasing analytic semigroup for real $t \geq 0$. For every positive time, the state has an entire extension in the finite complex control amplitudes.*

On $\mathcal{Q}_d^{\infty}$, the same statements hold on X_π for $q_0 \in X_\pi$, and the observable pairing is continuous for $V_w \in X_\pi^* \cap L^\infty$.

Proof. Put $q = \pi u$. Direct substitution using $b = \boldsymbol{D}\nabla \log \pi$ gives

$$\mathcal{L}(\pi u) = \pi \mathcal{G} u, \qquad \mathcal{G} = \pi^{-1}\nabla\cdot(\pi \boldsymbol{D}\nabla). \tag{S32}$$

For each fixed finite d, take the weighted H^1 form domain with periodic traces in all $d-1$ transverse coordinates and the declared weighted-Neumann convention on nonperiodic bounded faces. No Sobolev embedding or dimension-uniform estimate is used. On the weighted space $L^2(\pi\mathrm{d}x)$,

$$\langle u, \mathcal{G}v\rangle_{L^2(\pi)} = -\int (\nabla u)^{\mathsf{T}}\boldsymbol{D}\nabla v\,\pi\mathrm{d}x. \tag{S33}$$

The closed weighted-Neumann form therefore realizes $\mathcal{G}$ as a nonpositive self-adjoint generator. Its semigroup is positive and mass-conserving for real nonnegative time, and its norm is contractive for complex time with nonnegative real part. On $\mathcal{Q}_d^{\mathrm{box}}$, multiplication by π is a bounded isomorphism between the weighted and unweighted L^2 spaces. On $\mathcal{Q}_d^{\infty}$, the same map is unitary into X_π. Bounded multiplication by $-BV_w$ preserves the generator domain and analyticity. The Dyson–Phillips expansion proves entire finite-dimensional complex parameter dependence. Positivity and mass decrease follow from the product formula, or equivalently from Feynman–Kac, only for real nonnegative V_w, real $B \geq 0$, and real $t \geq 0$. $\square$

The proposition gives time regularization in powers of the generator, not global classical spatial regularity across the sharp contact interface. Initial boundary compatibility is unnecessary for positive-time mixed jets.

For the physical two-dimensional quotient used by the numerical program, the unbounded part of Proposition S3.1 has the following explicit consequence.

Corollary S3.2 (Physical natural-decay realization and observable calculus)**.** *Fix $d = 2$, $D, \gamma, W > 0$, $B \geq 0$, a real simplex control w, and the sharp bounded field V_w of Eq.* (S8)*. On*

$$H = L^2(\mathcal{Q}_d^\infty, \pi \mathrm{d}x), \qquad \mathcal{V} = \overline{C_c^\infty(\mathbb{R}^2) \otimes C^\infty(\mathbb{T}_W)}^{\|\cdot\|_{\mathcal{V}}}, \tag{S34}$$

where

$$\|u\|_{\mathcal{V}}^2 = \|u\|_H^2 + \int_{\mathcal{Q}_d^\infty} (\nabla u)^\mathsf{T} \boldsymbol{D} \nabla u \, \pi \mathrm{d}x, \tag{S35}$$

the killed form is a closed symmetric nonnegative Dirichlet form with this same domain. Its operator $H_{\infty,w} \geq 0$ generates a symmetric analytic sub-Markov contraction semigroup, and multiplication by π gives the unitary realization

$$A_{B,w} = U(-H_{\infty,w})U^{-1}, \qquad Uu = \pi u, \tag{S36}$$

of the natural-decay forward generator. Here

$$Z_\pi = \frac{2\pi D W}{\gamma}, \qquad \boldsymbol{D} \nabla \log \pi = b. \tag{S37}$$

For $u_0 = q_0/\pi \in H$, $r = 0, 1, 2$, and $t \in [\tau, T]$,

$$F_w^{(r)}(t) = (-1)^r \langle V_w, H_{\infty,w}^r \mathrm{e}^{-tH_{\infty,w}} u_0 \rangle_H, \tag{S38}$$

$$\sup_{t \in [\tau,T]} |F_w^{(r)}(t)| \leq \|V_w\|_H C_r(\tau) \|u_0\|_H, \quad C_0(\tau) = 1, \quad C_r(\tau) = \left(\frac{r}{e\tau}\right)^r \quad (r = 1, 2). \tag{S39}$$

If Eq. (S10) *holds, then $q_{B,w}(t) \geq 0$, $F_w(t) \geq 0$, and*

$$\int_{\mathcal{Q}_d^\infty} q_{B,w}(t) \mathrm{d}x + B \int_0^t F_w(s) \mathrm{d}s = 1. \tag{S40}$$

Proof. The two Gaussian integrals and the normalized torus factor give $Z_\pi = 2\pi D W/\gamma$, while direct differentiation gives $\boldsymbol{D} \nabla \log \pi = b$. Bounded nonnegative multiplication preserves the closed free form, its domain, and the Dirichlet property. The representation theorem and the Dirichlet-form correspondence give the asserted operator and semigroup, and Eq. (S36) follows by integration by parts, which identifies the form-associated realization with the displayed algebraic action on the core. No separate essential-self-adjointness assertion for the minimal core operator is used. The spectral theorem gives $\|H^r \mathrm{e}^{-tH}\| \leq \sup_{\lambda \geq 0} \lambda^r \mathrm{e}^{-\tau\lambda} = (r/(e\tau))^r$ for $r = 1, 2$, proving Eq. (S39). Gaussian cutoffs put the constant function one in $\mathcal{V}$. Testing the weak evolution with that function gives $\frac{\mathrm{d}}{\mathrm{d}t} \int q = -BF_w$; the sub-Markov property gives positivity, and integration proves Eq. (S40). □

Corollary S3.2 fixes the continuum operator realization only. It does not prove finite-volume Mosco convergence, a computable spatial or box-truncation error, or continuum stationary topology.

S3.2. A conditional varying-space positive-time transfer

The next proposition isolates the abstract implication that will be available if a model-specific finite-volume resolvent limit is proved. It is useful precisely because the reconstruction and projection maps need not be isometries at finite mesh width.

Proposition S3.3 (Varying-space positive-time functional bridge)**.** *Let $L_h \geq 0$ and $L \geq 0$ be self-adjoint operators on Hilbert spaces $\mathcal{H}_h$ and $\mathcal{H}$. Suppose that $J_h : \mathcal{H}_h \to \mathcal{H}$ and $P_h : \mathcal{H} \to \mathcal{H}_h$ satisfy*

$$P_h = J_h^*, \qquad \sup_h \big(\|J_h\| + \|P_h\|\big) < \infty, \qquad \delta_h := \|P_h J_h - I_{\mathcal{H}_h}\| \longrightarrow 0, \tag{S41}$$

and, for one $\alpha > 0$,

$$J_h (L_h + \alpha)^{-1} P_h u \longrightarrow (L + \alpha)^{-1} u \quad \text{strongly for every } u \in \mathcal{H}. \tag{S42}$$

Then, for $r = 0, 1, 2,$ $0 < \tau \leq T < \infty$, and every $u_0 \in \mathcal{H}$,

$$\sup_{t \in [\tau, T]} \left\| J_h L_h^r \mathrm{e}^{-tL_h} P_h u_0 - L^r \mathrm{e}^{-tL} u_0 \right\|_{\mathcal{H}} \longrightarrow 0. \tag{S43}$$

If $V_h \in \mathcal{H}_h$, $V \in \mathcal{H}$, and $J_h V_h \to V$ strongly, then also

$$\sup_{t \in [\tau, T]} \left| \left\langle V_h, L_h^r \mathrm{e}^{-tL_h} P_h u_0 \right\rangle_h - \left\langle V, L^r \mathrm{e}^{-tL} u_0 \right\rangle_{\mathcal{H}} \right| \longrightarrow 0. \tag{S44}$$

Proof. Put $R_h = (L_h + \alpha)^{-1}$, $R = (L + \alpha)^{-1}$, and $\widehat{R}_h = J_h R_h P_h$. Compression is not exactly multiplicative, but for every fixed $k \geq 2$ a telescoping expansion gives

$$\left\| \widehat{R}_h^k - J_h R_h^k P_h \right\| \leq \|J_h\| \|P_h\| (k-1) \alpha^{-k} (1 + \delta_h)^{k-2} \delta_h. \tag{S45}$$

Indeed, with $D_h = P_h J_h$, expand $R_h (D_h R_h)^{k-1} - R_h^k$ by replacing one D_h at a time. Equation (S42), uniform boundedness, and Eq. (S45) therefore give convergence for every polynomial in R_h with zero constant term.

For $f \in C_0([0, \infty))$, the change of variables $y = (\lambda + \alpha)^{-1}$ produces a continuous function on $[0, \alpha^{-1}]$ that vanishes at $y = 0$. Polynomials with zero constant term are uniformly dense in that class. The spectral theorem consequently gives

$$J_h f(L_h) P_h u \longrightarrow f(L) u \quad (u \in \mathcal{H}). \tag{S46}$$

This zero-constant approximation is why no false operator-norm assertion $J_h P_h \to I_{\mathcal{H}}$ is needed.

For $f_{r,t}(\lambda) = \lambda^r \mathrm{e}^{-t\lambda}$,

$$\|f_{r,t} - f_{r,s}\|_\infty \leq \left(\frac{r+1}{e\tau} \right)^{r+1} |t - s|. \tag{S47}$$

A finite time net, Eq. (S46) at its nodes, and Eq. (S47) between nodes prove Eq. (S43).

Finally, exact adjointness gives, for every $v_h \in \mathcal{H}_h$,

$$\begin{aligned} &\langle V_h, v_h \rangle_h - \langle V, J_h v_h \rangle_{\mathcal{H}} \\ &\quad = \langle V_h, (I - P_h J_h) v_h \rangle_h + \langle J_h V_h - V, J_h v_h \rangle_{\mathcal{H}}. \end{aligned} \tag{S48}$$

Moreover, $\|J_h V_h\|^2 = \langle V_h, P_h J_h V_h \rangle_h \geq (1 - \delta_h) \|V_h\|_h^2$ for small h, so $J_h V_h \to V$ implies $\sup_h \|V_h\|_h < \infty$. The spectral bound

$$\sup_{t \in [\tau, T]} \|L_h^r \mathrm{e}^{-tL_h} P_h u_0\|_h \leq C_r(\tau) \|P_h\| \|u_0\|, \qquad C_0 = 1, \quad C_r = (r/(e\tau))^r \tag{S49}$$

and Eq. (S43) now turn Eq. (S48) into Eq. (S44). □

Proposition S3.3 remains conditional in general. For the twelve formula-defined fixed-box dyadic families, a separate audited ideal composition now verifies its reconstructed-resolvent hypothesis and hence the stated Mosco and compact-positive-time conclusions. This closes an ideal fixed-box C1 composition only. It does not certify a correlated level-$n = 0$ production member, bind the production gauge/evaluator bridge, or provide a computable production error. Project-level and production-level complete C1, quantitative C2, continuum root margins, and release therefore remain false.

S3.3. The mixed-jet estimate

Define the budget-normalized density and exact free-exposure law by

$$F_B(t,\theta) = B^{-1} f_B(t, w(\theta)), \qquad G(t,\theta) = \langle V_{w(\theta)}, \mathrm{e}^{t\mathcal{L}} q_0 \rangle. \tag{S50}$$

The definition of F_B applies for $B > 0$; the theorem below defines its continuous extension at $B = 0$.

Let Θ be a compact real control set strictly inside the simplex. Fix $\delta > 0$ and use the enlarged complex tube

$$\Theta_\delta^+ = \left\{\zeta \in \mathbb{C}^{J-1} : \mathrm{dist}_\infty(\zeta, \Theta) < 2\delta\right\}. \tag{S51}$$

Every closed coordinate polydisc of radius δ about a point of Θ lies strictly inside Θ_δ^+; this is the Cauchy margin used below. Complex tube points are analytic auxiliaries and need not be nonnegative simplex weights. Set

$$v_{\infty,\delta} = \sup_{\zeta \in \Theta_\delta^+} \|V_{w(\zeta)}\|_\infty. \tag{S52}$$

For the bounded problem let

$$v_{*,\delta} = \sup_{\zeta \in \Theta_\delta^+} \|V_{w(\zeta)}\|_{L^2}, \qquad \kappa_\pi \le \sqrt{\pi_{\max}/\pi_{\min}}. \tag{S53}$$

For the unbounded problem replace the first norm by $\|\cdot\|_{X_\pi^*}$ and set $\kappa_\pi = 1$. In either case write $\|q_0\|_X$ for the corresponding state norm.

Theorem S3.4 (Uniform weak-reaction mixed-jet bridge)**.** *Fix $0 < \tau \le T < \infty$, $0 \le B \le B_{\max}$, an integer $r \ge 0$, and a finite control multi-index α. Then F_B extends continuously to $F_0 = G$, and*

$$\begin{aligned}
&\sup_{(t,\theta)\in[\tau,T]\times\Theta} |\partial_t^r \partial_\theta^\alpha (F_B - G)| \\
&\quad \le r!\,\alpha! \left(\frac{2}{\tau}\right)^r \delta^{-|\alpha|} v_{*,\delta} \kappa_\pi \left[\exp\left(\frac{3}{2} B v_{\infty,\delta} T\right) - 1\right] \|q_0\|_X.
\end{aligned} \tag{S54}$$

Here $\alpha! = \prod_i \alpha_i!$. Consequently the left side is at most $BC_{r,\alpha}$, where the explicit B-independent constant

$$\begin{aligned}
C_{r,\alpha} &= r!\,\alpha! \left(\frac{2}{\tau}\right)^r \delta^{-|\alpha|} v_{*,\delta} \kappa_\pi \frac{3T}{2} v_{\infty,\delta} \\
&\quad \times \exp\left(\frac{3}{2} B_{\max} v_{\infty,\delta} T\right) \|q_0\|_X
\end{aligned} \tag{S55}$$

works for all $0 \le B \le B_{\max}$.

Proof. In the self-adjoint representation, take complex time z with $\mathrm{Re}\, z > 0$, and define

$$\Delta_n = \{(s_1, \ldots, s_n) : 1 \ge s_1 \ge \cdots \ge s_n \ge 0\}. \tag{S56}$$

The norm-convergent Dyson expansion is

$$\mathrm{e}^{z(\mathcal{G} - BV_w)} = \mathrm{e}^{z\mathcal{G}} + \sum_{n=1}^\infty (-Bz)^n \int_{\Delta_n} \mathrm{e}^{z(1-s_1)\mathcal{G}} V_w \left[\prod_{k=1}^{n-1} \mathrm{e}^{z(s_k - s_{k+1})\mathcal{G}} V_w\right] \mathrm{e}^{z s_n \mathcal{G}}\, \mathrm{d}\boldsymbol{s}, \tag{S57}$$

where the product is ordered with increasing k from left to right and is the identity for $n = 1$; the ordered simplex has volume $1/n!$. Every free factor is a contraction because its scalar time increment has nonnegative real part. The norm estimates extend to $\mathrm{Re}\, z = 0$ by strong continuity, although the Cauchy disks below remain strictly inside $\mathrm{Re}\, z > 0$. Multiplication by V_w commutes with multiplication by π, so the similarity factor enters once rather than at every Dyson factor. Hence

$$\|\mathrm{e}^{zA_{B,w}} - \mathrm{e}^{z\mathcal{L}}\|_{X\to X} \le \kappa_\pi \{\mathrm{e}^{B|z|\|V_w\|_\infty} - 1\}. \tag{S58}$$

The complex-bilinear observable remainder

$$R_B(z,\zeta) = \langle V_{w(\zeta)}, [\mathrm{e}^{zA_{B,w(\zeta)}} - \mathrm{e}^{z\mathcal{L}}] q_0 \rangle \tag{S59}$$

therefore satisfies

$$|R_B(z,\zeta)| \le v_{*,\delta}\kappa_\pi\{\mathrm{e}^{Bv_{\infty,\delta}|z|}-1\}\|q_0\|_X. \tag{S60}$$

For $t\in[\tau,T]$, the disk $|z-t|\le\tau/2$ stays in the open right half-plane and obeys $|z|\le 3T/2$. Cauchy's estimate on that disk and on each closed radius-δ control polydisc contained in Θ_δ^+ gives Eq. (S54). Finally, $\mathrm{e}^{Bx}-1\le Bx\mathrm{e}^{B_{\max}x}$ gives Eq. (S55). □

Because G is affine in θ, Theorem S3.4 also gives

$$D_\theta F_B = D_\theta G + O(B), \qquad D_\theta^k F_B = O(B) \quad (k\ge 2), \tag{S61}$$

through every fixed number of time derivatives. This is an independent consistency check on the hierarchy in Sec. S2.

S3.4. First Dyson correction

For real time define

$$\mathcal{H}_w(t) = \int_0^t \langle V_w, \mathrm{e}^{(t-s)\mathcal{L}} M_{V_w} \mathrm{e}^{s\mathcal{L}} q_0\rangle\,\mathrm{d}s. \tag{S62}$$

Then

$$F_B = G - B\mathcal{H}_w + \mathcal{R}_{2,B}. \tag{S63}$$

Defining $\mathcal{R}_{2,B}$ by the $n\ge 2$ part of the complex Dyson series and applying the same Cauchy argument gives

$$\begin{aligned}&\sup_{[\tau,T]\times\Theta} |\partial_t^r\partial_\theta^\alpha \mathcal{R}_{2,B}| \\ &\le \frac{B^2}{2} r!\alpha! \left(\frac{2}{\tau}\right)^r \delta^{-|\alpha|} v_{*,\delta}\kappa_\pi \left(\frac{3}{2}v_{\infty,\delta}T\right)^2 \mathrm{e}^{(3/2)B_{\max}v_{\infty,\delta}T}\|q_0\|_X.\end{aligned} \tag{S64}$$

Thus the first correction has a genuine $O(B^2)$ remainder through every fixed mixed jet; a real-axis remainder bound alone would not justify the control derivatives in Eq. (S64).

S3.5. Quantitative local persistence, and what it does not imply

The mixed-jet estimate transfers strict finite collections of inequalities. For example, suppose on a closed interval $[a_-,a_+]\subset(\tau,T)$

$$G'(a_-) > \eta_1, \qquad G'(a_+) < -\eta_1, \qquad \sup_{[a_-,a_+]} G'' < -\eta_2 \tag{S65}$$

for $\eta_1,\eta_2>0$. If the C^2 mixed-jet error is smaller than $\min(\eta_1,\eta_2)$, then F_B has exactly one critical point in the interval and it is a nondegenerate maximum.

For folds and cusps, first freeze a square finite-dimensional unfolding. Let $H_B:\mathcal{U}\subset\mathbb{R}^n\to\mathbb{R}^n$ be the corresponding C^1 map, let $H_0(x_0)=0$, put $A_0=DH_0(x_0)$, and assume A_0 is invertible. Suppose the closed ball $\overline{B_r(x_0)}$ remains inside positive time and, after the control slice is mapped back to θ, the simplex interior. If

$$\sup_{x\in\overline{B_r(x_0)}} \|I-A_0^{-1}DH_B(x)\|_2 \le \frac{1}{2}, \tag{S66}$$

$$\|A_0^{-1}H_B(x_0)\|_2 \le \frac{r}{2}, \tag{S67}$$

then $x\mapsto x-A_0^{-1}H_B(x)$ is a contraction of the ball into itself. It has one zero x_B in that ball and

$$\|x_B-x_0\|_2 \le 2\|A_0^{-1}H_B(x_0)\|_2. \tag{S68}$$

For a fold, freeze one unit vector $e \in \mathbb{R}^{J-1}$ in the dimensionless θ-metric, set $\theta = \bar\theta + e\lambda$, and use the square map

$$H_B^{\mathrm{fold}}(t,\lambda) = \big((F_B)_t, (F_B)_{tt}\big)(t, \bar\theta + e\lambda) \in \mathbb{R}^2. \tag{S69}$$

For a cusp, freeze an isometric two-plane

$$E \in \mathbb{R}^{(J-1)\times 2}, \qquad E^{\mathsf{T}} E = I_2, \qquad \theta = \bar\theta + E\xi, \tag{S70}$$

in the same metric, and use

$$H_B^{\mathrm{cusp}}(t,\xi) = \big((F_B)_t, (F_B)_{tt}, (F_B)_{ttt}\big)(t, \bar\theta + E\xi) \in \mathbb{R}^3. \tag{S71}$$

The vectors e and the columns of E, as well as the nondimensionalization, are frozen before B is varied.

Define the full normalized response and the response restricted to that same cusp plane by

$$R_B(x) = \begin{pmatrix} \nabla_\theta (F_B)_t(x) \\ \nabla_\theta (F_B)_{tt}(x) \end{pmatrix}, \qquad R_B^E(x) = R_B(x)E \in \mathbb{R}^{2\times 2}. \tag{S72}$$

where $x = (t,\xi)$ means evaluation at $(t, \bar\theta + E\xi)$. At a cusp root, with columns ordered as (t, ξ_1, ξ_2),

$$\det DH_B^{\mathrm{cusp}} = (F_B)_{tttt} \det R_B^E. \tag{S73}$$

Thus the quartic time derivative must be nonzero and the restricted two-control response must be nonsingular. If on the same cusp ball

$$s_* := \inf_x \sigma_2(R_0^E(x)) > 0, \qquad \varepsilon_R(B) := \sup_x \|R_B^E(x) - R_0^E(x)\|_2 < s_*, \tag{S74}$$

then Weyl's inequality gives

$$\inf_x \sigma_2(R_B^E(x)) \ge s_* - \varepsilon_R(B) > 0, \tag{S75}$$

including at the displaced root. This rank-two condition applies only when two independent budget tangents exist. The raw response for f_B is BR_B^E, so normalized rank may persist while both its absolute singular value and event rate vanish as $B \downarrow 0$.

Theorem S3.4 is local in positive time. It does not exclude late or remote extra extrema, provide a finite event-mass floor, prove that a specified finite B lies in its asymptotic regime, or certify a numerical discretization. Every application requires explicit margins such as Eqs. (S65), (S66), and (S74).

S4. CONSTRUCTIVE FIXED-FINITE-MODE THEOREM IN EVERY FIXED FINITE $d \geq 2$

S4.1. Narrow-noise OU slab family and initial law

Fix once and for all

$$d \in \mathbb{N},\ d \ge 2, \quad D_0, \gamma, W, \rho, s_0, u_0 > 0, \quad 0 < a < W/2, \quad z_0 \neq \bar z. \tag{S76}$$

For $\epsilon > 0$, let the longitudinal midpoint solve

$$\mathrm{d}Z_t = -\gamma(Z_t - \bar z)\,\mathrm{d}t + \epsilon\sqrt{D_0}\,\mathrm{d}W_t, \qquad Z_0 \sim N(z_0, \epsilon^2 s_0^2). \tag{S77}$$

For each fixed ϵ, this is the generic model of Sec. S1 with the explicit identification $D = \epsilon^2 D_0$; the same identification gives the relative-coordinate noise amplitudes below. Its mean and variance are

$$\mu(t) = \bar z + (z_0 - \bar z)\mathrm{e}^{-\gamma t}, \tag{S78}$$

$$\operatorname{Var} Z_t = \epsilon^2 s^2(t), \qquad s^2(t) = s_0^2 \mathrm{e}^{-2\gamma t} + \frac{D_0}{2\gamma}(1 - \mathrm{e}^{-2\gamma t}). \tag{S79}$$

The relative quotient follows

$$\mathrm{d}R_{\parallel,t} = -\gamma R_{\parallel,t}\,\mathrm{d}t + 2\epsilon\sqrt{D_0}\,\mathrm{d}W_{\parallel,t}, \tag{S80}$$

$$\mathrm{d}R_{\perp,t} = 2\epsilon\sqrt{D_0}\,\mathrm{d}\boldsymbol{W}_{\perp,t} \quad (\mathrm{mod}\ W). \tag{S81}$$

The complete initial law is the mutually independent product

$$R_{\|,0} \sim N(r_{\|,0}, \epsilon^2 u_0^2), \tag{S82}$$

$$R_{\perp,0} \sim \mathrm{WN}(r_{\perp,0}, \epsilon^2 \Sigma_{\perp,0}), \qquad \Sigma_{\perp,0} \succ 0, \tag{S83}$$

and all initial variables and Brownian drivers are mutually independent. Here $r_{\perp,0} \in \mathbb{T}_W^{d-1}$ and $\Sigma_{\perp,0} \in \mathbb{R}^{(d-1)\times(d-1)}$ is positive definite.

For each fixed $\epsilon > 0$, the invariant density π_ϵ is Gaussian with mean $\bar{z}$ and variance $\epsilon^2 D_0/(2\gamma)$ in Z, Gaussian with mean zero and variance $2\epsilon^2 D_0/\gamma$ in $R_\|$, and uniform on the transverse torus. Direct Gaussian exponent comparison gives

$$q_0 \in X_{\pi_\epsilon} \quad \Longleftrightarrow \quad s_0^2 < \frac{D_0}{\gamma} \text{ and } u_0^2 < \frac{4D_0}{\gamma} \tag{S84}$$

for the two unbounded Gaussian factors; the wrapped factor is square integrable against the uniform torus measure. We impose both strict inequalities. They hold pointwise in ϵ, but the weighted norm is not asserted to remain bounded as $\epsilon \downarrow 0$.

For fixed $\rho > 0$, define the normalized slabs

$$\phi_{j,\epsilon}(z) = \frac{1}{\sqrt{2\pi}\epsilon\rho} \exp\left[-\frac{(z-c_j)^2}{2\epsilon^2\rho^2}\right] \tag{S85}$$

and the killing field

$$K_{B,w,\epsilon}(Z,R) = \frac{B}{W^{d-1}} \chi_a(R) \sum_{j=1}^m w_j \phi_{j,\epsilon}(Z), \qquad w_j \geq 0, \quad \sum_j w_j = 1. \tag{S86}$$

Each slab has unit full centre-space integral, so the installed amount remains B. The catalyst is a longitudinal slab, not an arbitrary localized two- or three-dimensional patch.

S4.2. Target times and the contact-tail lemma

Fix $0 < \tau < T < \infty$ and distinct target times

$$\tau < t_1 < \cdots < t_m < T. \tag{S87}$$

Because $z_0 \neq \bar{z}$, μ is strictly monotone on this window. Choose pairwise disjoint closed target neighborhoods whose union is I_*, with $I_* \subset (\tau, T)$. The deterministic relative path is

$$r_*(t) = \big(r_{\|,0}\mathrm{e}^{-\gamma t}, r_{\perp,0}\big). \tag{S88}$$

Assume a fixed contact-interior margin

$$\sup_{t\in I_*} |r_*(t)|_{\mathrm{mi}} \leq a - \eta \quad \text{for some } \eta > 0. \tag{S89}$$

Finally set $c_j = \mu(t_j)$.

Let $c_{d,\epsilon}(t) = \Pr\{|R_t|_{\mathrm{mi}} < a\}$. The relative covariance is $\epsilon^2 \Sigma_R(t)$, where its longitudinal coefficient and its unwrapped transverse coefficient are

$$u_0^2 \mathrm{e}^{-2\gamma t} + \frac{2D_0}{\gamma}(1 - \mathrm{e}^{-2\gamma t}), \tag{S90}$$

$$\Sigma_{\perp,0} + 4D_0 t I_{d-1}, \tag{S91}$$

respectively.

Lemma S4.1 (Differentiated contact tail)**.** *For every fixed integer $r \geq 0$, there exist finite constants $C_{r,d}, N_{r,d}, q_d > 0$, independent of all sufficiently small ϵ, such that*

$$\sup_{t\in I_*} |\partial_t^r \{c_{d,\epsilon}(t) - 1\}| \leq C_{r,d}\epsilon^{-N_{r,d}} \mathrm{e}^{-q_d/\epsilon^2}. \tag{S92}$$

Proof. Let $C_a = \{R : |R|_{\mathrm{mi}} < a\}$, and use the product geodesic distance on $\mathbb{R} \times \mathbb{T}_W^{d-1}$. The reverse triangle inequality and Eq. (S89) give the chart-independent bound

$$\inf_{t\in I_*} \inf_{R\in C_a^c} d_{\mathrm{cyl}}(R, r_*(t)) \geq a - \sup_{t\in I_*} |r_*(t)|_{\mathrm{mi}} \geq \eta. \tag{S93}$$

The condition $a < W/2$ keeps the contact boundary away from the torus cut locus. Uniform positive definiteness and boundedness of the covariance and its fixed time derivatives give constants

$$0 < \lambda_- \leq \lambda_+ < \infty, \qquad \lambda_- I \preceq \Sigma_R(t) \preceq \lambda_+ I, \qquad t \in I_*. \tag{S94}$$

Let $p_{\epsilon,t}$ be the normalized relative density and write its wrapped transverse part as a lattice-image sum. If $Y_n(R,t)$ is the Euclidean displacement from the mean in image $n \in \mathbb{Z}^{d-1}$, differentiation of one Gaussian image gives, for fixed r, integers M_r, N_r, and constants independent of small ϵ,

$$\left|\partial_t^r p_{\epsilon,t}^{(n)}(R)\right| \leq C_r \epsilon^{-d-N_r} \left[1 + \left(\frac{|Y_n(R,t)|}{\epsilon}\right)^{M_r}\right] \exp\left[-\frac{|Y_n(R,t)|^2}{C_r \epsilon^2}\right]. \tag{S95}$$

Uniform Gaussian summability on the compact positive-time set justifies termwise differentiation and summation. Equation (S93) means that every Euclidean lift, including whichever image is nearest a fundamental-cell face, has $|Y_n| \geq \eta$ on C_a^c. More explicitly, partition $\mathbb{Z}^{d-1}$ into max-norm shells. The number of indices in shell k is at most $(2k+3)^{d-1} - (2k+1)^{d-1}$, whereas periodicity and the fixed fundamental cell give $|Y_n| \geq c_d k - C_d$. Thus the polynomial factor in Eq. (S95) is dominated by a summable fixed-dimensional Gaussian shell majorant, uniformly for small ϵ.

Finally, normalization supplies the step needed to estimate derivatives by a tail integral:

$$\int \partial_t^r p_{\epsilon,t}(R)\,\mathrm{d}R = 0 \quad (r \geq 1), \qquad \partial_t^r (c_{d,\epsilon} - 1) = -\int_{C_a^c} \partial_t^r p_{\epsilon,t}(R)\,\mathrm{d}R \quad (r \geq 0). \tag{S96}$$

For $r = 0$, the first equality is replaced by $c_{d,\epsilon} - 1 = -\int_{C_a^c} p_{\epsilon,t}$. Integrating Eq. (S95) on this uniformly separated set and summing the images gives a polynomial prefactor times $\exp(-q/\epsilon^2)$, proving Eq. (S92). ☐

In particular, $c_{d,\epsilon} \to 1$ in $C^2(I_*)$. The proof uses the contact-interior hypothesis essentially; arbitrary approach or separation paths are not covered.

S4.3. Exact free clocks and channel separation

Independence of midpoint and relative motion gives the exact free-exposure channel

$$g_{j,\epsilon}(t) = \frac{c_{d,\epsilon}(t)}{W^{d-1}\sqrt{2\pi}\epsilon S(t)} \exp\left[-\frac{(c_j - \mu(t))^2}{2\epsilon^2 S^2(t)}\right], \tag{S97}$$

$$S^2(t) = s^2(t) + \rho^2, \tag{S98}$$

and

$$G_\epsilon(t; w) = \sum_{j=1}^m w_j g_{j,\epsilon}(t). \tag{S99}$$

Lemma S4.2 (Own-channel local Gaussian)**.** *For fixed $L < \infty$, uniformly on $|y| \leq L$ and for $r = 0, 1, 2$,*

$$\epsilon^{r+1} \partial_t^r g_{j,\epsilon}(t_j + \epsilon y) \longrightarrow \partial_y^r A_j(y), \tag{S100}$$

$$A_j(y) = \frac{1}{W^{d-1}\sqrt{2\pi} S(t_j)} \exp\left[-\frac{\mu'(t_j)^2 y^2}{2S^2(t_j)}\right]. \tag{S101}$$

There is one common $L_0 > 0$, independent of small ϵ, such that on

$$I_{j,\epsilon} = [t_j - L_0\epsilon, t_j + L_0\epsilon] \tag{S102}$$

the own-channel endpoint slopes have signs $+, -$ and magnitude $O(\epsilon^{-2})$, while the curvature is strictly negative with magnitude $O(\epsilon^{-3})$. For all sufficiently small ϵ, these intervals are pairwise disjoint and each is contained in its declared component of I_*.

Proof. Taylor expansion on a fixed y-interval gives

$$\frac{c_j - \mu(t_j + \epsilon y)}{\epsilon} = -\mu'(t_j)y + O(\epsilon) \tag{S103}$$

in C^2, and $S(t_j + \epsilon y) \to S(t_j) > 0$ in the same topology. Smooth composition in Eq. (S97), followed by Lemma S4.1, proves Eq. (S100). Since there are only finitely many channels and

$$A_j''(y) = \frac{\mu'(t_j)^2}{S(t_j)^2}\left(\frac{\mu'(t_j)^2 y^2}{S(t_j)^2} - 1\right) A_j(y), \tag{S104}$$

choose $0 < L_0 < \min_j S(t_j)/|\mu'(t_j)|$, decreasing it if needed to fit the declared target neighborhoods. Every limiting Gaussian then has positive left slope, negative right slope, and strictly negative curvature. Uniform C^2 convergence then gives constants $\alpha_j, \kappa_j > 0$ such that

$$\begin{aligned} \partial_t g_{j,\epsilon}(t_j - L_0\epsilon) &\geq \alpha_j \epsilon^{-2}, \\ \partial_t g_{j,\epsilon}(t_j + L_0\epsilon) &\leq -\alpha_j \epsilon^{-2}, \\ \sup_{I_{j,\epsilon}} \partial_t^2 g_{j,\epsilon} &\leq -\kappa_j \epsilon^{-3}. \end{aligned} \tag{S105}$$

□

Lemma S4.3 (Cross-channel exponential bound)**.** *For $i \neq j$, all sufficiently small ϵ, and $r = 0, 1, 2$,*

$$\sup_{t \in I_{j,\epsilon}} |\partial_t^r g_{i,\epsilon}(t)| \leq C_{ijr} \epsilon^{-M_r} \mathrm{e}^{-q_{ij}/\epsilon^2} \tag{S106}$$

for constants independent of ϵ.

Proof. Strict monotonicity gives $c_i \neq c_j$. On the shrinking jth interval, $|c_i - \mu(t)| \geq |c_i - c_j|/2$ for all sufficiently small ϵ. Every fixed time derivative of Eq. (S97) adds only a polynomial power of ϵ^{-1}; the separated Gaussian exponent remains $-q_{ij}/\epsilon^2$. Lemma S4.1 controls the common contact factor and its derivatives, proving Eq. (S106). □

S4.4. The theorem and its quantifier order

Let

$$\mathcal{W} \subset \left\{ w \in \mathbb{R}^m : w_j \geq w_{\min} > 0,\ \sum_j w_j = 1 \right\} \tag{S107}$$

be compact and nonempty, so necessarily $w_{\min} \leq 1/m$.

Theorem S4.4 (At least m modes in every fixed finite dimension)**.** *Fix integers $d \geq 2$ and $m \geq 1$, all dimension-sized data in Eqs.* (S76), (S82), *and* (S83), *the strict weighted-space conditions* (S84), *target times* (S87), *the contact-interior condition* (S89), *and the compact interior weight set $\mathcal{W}$. Then*

$$\boxed{\exists \epsilon_0(d, m, \mathit{data}, \mathcal{W}) > 0\ \ \forall \epsilon \in (0, \epsilon_0)\ \exists B_0(d, m, \mathit{data}, \mathcal{W}, \epsilon) > 0\ \ \forall B \in (0, B_0)\ \forall w \in \mathcal{W}:\ \mathsf{P}_m(B, \epsilon, w),} \tag{S108}$$

where P_m means that the exact continuum Doi density has exactly one nondegenerate local maximum in each of the named disjoint intervals $I_{j,\epsilon}$, and hence at least m local maxima. It also has at least one local minimum between every consecutive pair of certified intervals. No exact global root count is asserted.

Proof. Fix j. The own-channel contribution to the endpoint slopes is at least $w_{\min}\alpha_j\epsilon^{-2}$, and its negative-curvature margin is at least $w_{\min}\kappa_j\epsilon^{-3}$. Because m is fixed and $0 \leq w_i \leq 1$, Lemma S4.3 implies that the sum of all cross-channel first and second derivatives is smaller than half of every own-channel margin once ϵ is sufficiently small. The same ϵ_0 works for all finitely many intervals and all $w \in \mathcal{W}$. Thus

$$\begin{aligned} G_\epsilon'(t_j - L_0\epsilon; w) &> 0, \\ G_\epsilon'(t_j + L_0\epsilon; w) &< 0, \\ G_\epsilon''(t; w) &< 0 \quad (t \in I_{j,\epsilon}). \end{aligned} \tag{S109}$$

The derivative is strictly decreasing and changes sign exactly once, so the interval contains one unique nondegenerate maximum. On the gap between two consecutive intervals, the derivative is negative at the left endpoint and positive at the right endpoint. A minimum of the continuous function on the closed gap cannot occur at either endpoint, hence at least one local minimum is interior. It need not be nondegenerate.

To connect this weight set to Sec. S3, choose $\bar{w} \in \mathcal{W}$ and a frozen matrix $P \in \mathbb{R}^{m\times(m-1)}$ whose columns span $\{h : \mathbf{1}^{\mathsf{T}}h = 0\}$. Every $w \in \mathcal{W}$ has a unique coordinate $w = \bar{w} + P\theta$; hence

$$\Theta_{\mathcal{W}} = \{\theta : \bar{w} + P\theta \in \mathcal{W}\} \tag{S110}$$

is compact and lies strictly inside the real simplex chart. Choose the enlarged complex tube of Eq. (S51) around this set. Its complex points need not be probability weights: they are used only for the entire affine-control extension and Cauchy estimates.

Now fix one $\epsilon \in (0, \epsilon_0)$. Every Gaussian slab is then bounded, the initial law belongs to X_{π_ϵ}, and all tube norms are finite. The unbounded version of Theorem S3.4, with constants allowed to depend on this fixed ϵ, gives

$$\frac{f_{B,\epsilon}}{B} \longrightarrow G_\epsilon \quad \text{in } C^2([\tau, T]), \tag{S111}$$

uniformly over $\mathcal{W}$ as $B \downarrow 0$. The finitely many strict inequalities in Eq. (S109) therefore persist for all $0 < B < B_0(\epsilon)$, for one $B_0(\epsilon) > 0$ uniform over $\mathcal{W}$. The same sign and concavity argument proves P_m. □

The order in Eq. (S108) is essential. The semigroup constants can deteriorate rapidly as $\epsilon \downarrow 0$, because the profiles grow like ϵ^{-1} and the weighted initial norm need not be uniform. There is no interchange of the ϵ and B limits.

Corollary S4.5 (Leading peak balance and event-mass boundary)**.** *Let*

$$H_j = \frac{1}{W^{d-1}\sqrt{2\pi}S(t_j)}, \qquad w_j^{\mathrm{bal}} = \frac{S(t_j)}{\sum_i S(t_i)}. \tag{S112}$$

Apply Theorem S4.4 with the compact singleton $\mathcal{W} = \{w^{\mathrm{bal}}\}$. Then the leading certified free-exposure peak heights are equal. Denote the unique free-exposure maximum in $I_{j,\epsilon}$ by $t_{j,\epsilon}^{G,*}$. Then

$$\frac{t_{j,\epsilon}^{G,*} - t_j}{\epsilon} \to 0, \qquad \epsilon G_\epsilon(t_{j,\epsilon}^{G,*}; w^{\mathrm{bal}}) \to w_j^{\mathrm{bal}} H_j. \tag{S113}$$

Moreover,

$$\int_{I_{j,\epsilon}} G_\epsilon(t; w^{\mathrm{bal}})\,\mathrm{d}t \longrightarrow w_j^{\mathrm{bal}} \int_{-L_0}^{L_0} A_j(y)\,\mathrm{d}y > 0, \tag{S114}$$

whereas, after fixing ϵ, the physical Doi event mass satisfies, as $B \downarrow 0$,

$$\int_{I_{j,\epsilon}} f_{B,\epsilon}(t; w^{\mathrm{bal}})\,\mathrm{d}t = B \int_{I_{j,\epsilon}} G_\epsilon(t; w^{\mathrm{bal}})\,\mathrm{d}t + O_\epsilon(B^2). \tag{S115}$$

The positive-B Doi maximum may separately be denoted $t_{j,\epsilon}^{B,*}$; Eq. (S113) *makes no assertion that interchanges or couples the $\epsilon \downarrow 0$ and $B \downarrow 0$ limits. Thus the theorem supplies no positive absolute event-mass floor.*

Proof. Lemma S4.2, strict local concavity, and exponentially small cross channels locate the free-exposure maximum at $y = o(1)$ and give Eq. (S113). Changing variables $t = t_j + \epsilon y$ proves Eq. (S114). The first-order weak-reaction expansion in Eq. (S63), integrated over the interval after ϵ has been fixed, gives Eq. (S115); its remainder constant is allowed to depend on that fixed ϵ. □

Theorem S4.4 is pointwise in the fixed finite integers d and m, and is an existence result for a d-, m-, and ϵ-dependent slab family. No constant, dimensional budget, amplitude, event mass, or value of B_0 is uniform or compared across dimensions. It does not show that one fixed geometry realizes arbitrary m, exclude extra early, late, or interstitial extrema, certify nondegenerate separator minima, cover arbitrary localized patches, take a $d \to \infty$ limit, or demonstrate that nontrivial approach/separation dynamics causes the peaks. The latter limitation is visible in Lemma S4.1: the construction makes the contact factor tend to one near every designed clock.

The preceding theorem is retained because it allows a time-varying midpoint variance and local contact assumptions. It does not support the exact-mode headline. The stricter stationary-midpoint, whole-window subfamily needed for that headline is proved next.

S5. EXACT PRESCRIBED FINITE MODALITY: COMPLETE PROOF

This section proves the exact finite-mode statement used in the theorem-first working paper. The construction is pointwise in a fixed finite dimension and a fixed finite mode count. Its two small parameters are ordered: first the narrow-noise parameter is fixed sufficiently small, and only then is the Doi budget taken sufficiently small. All stationary-point counts below are on the declared compact positive-time window.

S5.1. Exact quotient, stationary midpoint variance, and whole-window contact

Fix finite integers $d \geq 2$ and $m \geq 1$, a transverse period $W > 0$, a contact radius $0 < a < W/2$, and a compact positive-time interval $I = [\tau, T] \subset (0, \infty)$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$, let the longitudinal midpoint and relative coordinates solve

$$\mathrm{d}Z_t = -\gamma(Z_t - \bar{z})\,\mathrm{d}t + \varepsilon\sqrt{D_0}\,\mathrm{d}W_t, \tag{S116}$$

$$\mathrm{d}R_{\parallel,t} = -\gamma R_{\parallel,t}\,\mathrm{d}t + 2\varepsilon\sqrt{D_0}\,\mathrm{d}W_{\parallel,t}, \tag{S117}$$

$$\mathrm{d}R_{\perp,t} = 2\varepsilon\sqrt{D_0}\,\mathrm{d}W_{\perp,t} \pmod W, \tag{S118}$$

where $D_0, \gamma > 0$. All Brownian drivers and all initial variables are mutually independent, and

$$Z_0 \sim N\left(z_0, \varepsilon^2 \frac{D_0}{2\gamma}\right), \qquad z_0 \neq \bar{z}, \tag{S119}$$

$$R_{\parallel,0} \sim N(r_{\parallel,0}, \varepsilon^2 u_0^2), \qquad 0 < u_0^2 < \frac{4D_0}{\gamma}, \tag{S120}$$

$$R_{\perp,0} \sim \mathrm{WN}(r_{\perp,0}, \varepsilon^2 \Sigma_{\perp,0}), \qquad \Sigma_{\perp,0} \succ 0. \tag{S121}$$

Here WN denotes the wrapped Gaussian on the transverse torus. The two strict variance inequalities $D_0/(2\gamma) < D_0/\gamma$ and $u_0^2 < 4D_0/\gamma$ are precisely the unbounded Gaussian conditions that place this initial law in the weighted state space used by Theorem S3.4; the wrapped factor is square integrable against the uniform torus measure.

The midpoint has mean and variance

$$\mu(t) = \bar{z} + (z_0 - \bar{z})\mathrm{e}^{-\gamma t}, \qquad \operatorname{Var} Z_t = \varepsilon^2 \frac{D_0}{2\gamma} \quad (t \geq 0). \tag{S122}$$

Thus only the variance, not the mean, is stationary. This constant variance is what produces a common-variance Gaussian mixture below.

Let $\varsigma = \operatorname{sgn} \mu' \in \{-1, 1\}$, fix a physical reference length $\ell_0 > 0$, and define

$$x(t) = \frac{\varsigma\mu(t)}{\ell_0}, \qquad v_0 := \inf_{t \in I} x'(t) > 0. \tag{S123}$$

Choose target times and their dimensionless centres by

$$\tau < t_1 < \cdots < t_m < T, \qquad c_j = x(t_j). \tag{S124}$$

Then $c_1 < \cdots < c_m$, and the physical centre of slab j is $\varsigma\ell_0 c_j = \mu(t_j)$. In particular, reversing the trajectory orientation reverses the physical centres at the same time; no centre is held fixed under the coordinate reversal.

Write

$$r_*(t) = (r_{\parallel,0}\mathrm{e}^{-\gamma t}, r_{\perp,0}) \tag{S125}$$

and impose the whole-window, minimum-image contact margin

$$\sup_{t \in I} |r_*(t)|_{\mathrm{mi}} \leq a - \eta \quad \text{for some } \eta > 0. \tag{S126}$$

The word "whole-window" is essential: the condition is required on every point of I, not merely near the target times.

Lemma S5.1 (Whole-window differentiated contact tail)**.** *Let*

$$c_{d,\varepsilon}(t) = \mathbb{P}\{|R_t|_{\mathrm{mi}} < a\}. \tag{S127}$$

Under Eqs. (S117)– (S126)*, for $r = 0, 1, 2$ there are finite constants $C_{r,d}, N_{r,d}, q_d > 0$, independent of all sufficiently small ε, such that*

$$\|\partial_t^r (c_{d,\varepsilon} - 1)\|_{L^\infty(I)} \le C_{r,d}\varepsilon^{-N_{r,d}} \exp(-q_d/\varepsilon^2). \tag{S128}$$

Consequently, $c_{d,\varepsilon} \ge 1/2$ for sufficiently small ε, and the first two logarithmic derivatives of $c_{d,\varepsilon}$ are uniformly bounded on I and converge to zero faster than any power of ε.

Proof. In an unwrapped transverse chart the relative Gaussian has mean $r_*(t)$ and covariance $\varepsilon^2\Sigma_R(t)$, with longitudinal and transverse blocks

$$u_0^2 \mathrm{e}^{-2\gamma t} + \frac{2D_0}{\gamma}(1 - \mathrm{e}^{-2\gamma t}), \qquad \Sigma_{\perp,0} + 4D_0 t I_{d-1}. \tag{S129}$$

Because $I \subset (0, \infty)$, $u_0 > 0$, and $\Sigma_{\perp,0} \succ 0$, this covariance and its first two time derivatives are uniformly bounded on I, and its eigenvalues lie between two positive constants. The reverse triangle inequality and Eq. (S126) give

$$\inf_{t\in I} \inf_{|r|_{\mathrm{mi}} \ge a} d_{\mathrm{cyl}}(r, r_*(t)) \ge \eta. \tag{S130}$$

The assumption $a < W/2$ keeps the contact ball away from the transverse cut locus. Express the wrapped density as its Gaussian lattice-image sum. Every time derivative of order at most two is the same Gaussian multiplied by a polynomial in the scaled displacement and in ε^{-1}, with degree bounded independently of t. Equation (S130), the uniform covariance bounds, and a standard Gaussian tail estimate therefore give, after summing the lattice images,

$$\sup_{t\in I} \int_{|r|_{\mathrm{mi}} \ge a} |\partial_t^r p_{\varepsilon,t}(r)|\, \mathrm{d}r \le C_{r,d}\varepsilon^{-N_{r,d}} \mathrm{e}^{-q_d/\varepsilon^2}, \qquad r = 0, 1, 2. \tag{S131}$$

Differentiation under the integral is justified by the same integrable majorant and yields Eq. (S128). For small ε, its $r = 0$ case gives $c_{d,\varepsilon} \ge 1/2$, while

$$\begin{aligned} |\partial_t \log c_{d,\varepsilon}| &\le 2|\partial_t c_{d,\varepsilon}|, \\ |\partial_t^2 \log c_{d,\varepsilon}| &\le 2|\partial_t^2 c_{d,\varepsilon}| + 4|\partial_t c_{d,\varepsilon}|^2. \end{aligned} \tag{S132}$$

This proves the logarithmic-derivative statement. □

Fix $\rho > 0$, put

$$S_*^2 = \frac{D_0}{2\gamma} + \rho^2, \qquad \sigma = \frac{\varepsilon S_*}{\ell_0}, \tag{S133}$$

and use the normalized physical slabs

$$\phi_{j,\varepsilon}(z) = \frac{1}{\sqrt{2\pi}\,\varepsilon\rho} \exp\left[-\frac{(z - \mu(t_j))^2}{2\varepsilon^2\rho^2}\right]. \tag{S134}$$

Let $0 < w_* \le 1/m$ and

$$\mathcal{W}_{w_*} = \left\{ w = (w_1, \ldots, w_m) : \sum_{j=1}^m w_j = 1, \quad w_j \ge w_* > 0 \right\}. \tag{S135}$$

For $w \in \mathcal{W}_{w_*}$, define the Doi killing field

$$K_{B,w,\varepsilon}(Z, R) = \mathbf{1}_{\{|R|_{\mathrm{mi}} < a\}} \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\varepsilon}(Z) = BV_{w,\varepsilon}(Z, R). \tag{S136}$$

Every slab has unit full longitudinal integral, so B is the same installed centre-space budget for every allocation.

Lemma S5.2 (Exact common-variance free-exposure factorization)**.** *Let $T_0(t)$ be the unkilled semigroup and q_0 the initial law above. The unit-budget free-exposure clock is exactly*

$$G_{\varepsilon,w}(t) := \langle V_{w,\varepsilon}, T_0(t)q_0\rangle = a_\varepsilon(t)H_{\sigma,w}(x(t)), \tag{S137}$$

$$a_\varepsilon(t) := \frac{c_{d,\varepsilon}(t)}{W^{d-1}\sqrt{2\pi}\,\varepsilon S_*}, \tag{S138}$$

$$H_{\sigma,w}(x) := \sum_{j=1}^{m} w_j \exp\left[-\frac{(x-c_j)^2}{2\sigma^2}\right]. \tag{S139}$$

The positive factor a_ε has uniformly bounded first two logarithmic time derivatives for all sufficiently small ε.

Proof. Independence factors the contact event from every longitudinal slab expectation. By Eq. (S122), convolution of $\phi_{j,\varepsilon}$ with the law of Z_t gives

$$\mathbb{E}[\phi_{j,\varepsilon}(Z_t)] = \frac{1}{\sqrt{2\pi}\,\varepsilon S_*}\exp\left[-\frac{(\mu(t)-\mu(t_j))^2}{2\varepsilon^2 S_*^2}\right]. \tag{S140}$$

Since $\mu(t)-\mu(t_j) = \varsigma\ell_0(x(t)-c_j)$, Eqs. (S133) and (S140) give Eqs. (S137)– (S139). The last assertion follows from Lemma S5.1, because the remaining factors in a_ε are independent of time. ☐

S5.2. The pure common-variance mixture

Set $J = x(I) = [\alpha,\beta]$. The targets are interior, so

$$\alpha < c_1 < \cdots < c_m < \beta. \tag{S141}$$

For $m \geq 2$, let

$$\Delta_j = c_{j+1} - c_j, \qquad \Delta_* := \min_{1\leq j<m} \Delta_j > 0. \tag{S142}$$

No empty minimum is used when $m = 1$.

Define

$$q_j(x) = w_j \mathrm{e}^{-(x-c_j)^2/(2\sigma^2)}, \qquad \pi_j(x) = \frac{q_j(x)}{H_{\sigma,w}(x)}, \qquad \bar{c}(x) = \sum_{j=1}^{m}\pi_j(x)c_j. \tag{S143}$$

The logarithmic slope $L_{\sigma,w} = \partial_x \log H_{\sigma,w}$ obeys the exact identities

$$L_{\sigma,w}(x) = \frac{\bar{c}(x)-x}{\sigma^2}, \tag{S144}$$

$$L'_{\sigma,w}(x) = \frac{\mathrm{Var}_{\pi(x)}(c)}{\sigma^4} - \frac{1}{\sigma^2}. \tag{S145}$$

Indeed, differentiation of q_j gives the first identity, while $\pi_j' = \pi_j(c_j-\bar{c})/\sigma^2$ gives the second.

The mode cap for univariate Gaussian mixtures is classical [4, 5]. We retain the direct multiplicity-counting proof because the later Doi transfer also needs explicit uniform root locations, curvature scales, and full-complement margins.

Lemma S5.3 (Global extended-Chebyshev zero bound)**.** *For distinct real centres and positive weights, $H'_{\sigma,w}$ has at most $2m-1$ real zeros counted with multiplicity. Its real zero set is finite.*

Proof. Multiplication by a nowhere-zero function preserves both zeros and their multiplicities, and direct differentiation gives

$$\sigma^2\mathrm{e}^{x^2/(2\sigma^2)}H'_{\sigma,w}(x) = \sum_{j=1}^{m} w_j(c_j-x)\mathrm{e}^{-c_j^2/(2\sigma^2)}\mathrm{e}^{c_jx/\sigma^2}. \tag{S146}$$

We prove the slightly more general claim that a nonzero function

$$P_m(x) = \sum_{j=1}^{m}(a_j+b_jx)\mathrm{e}^{\lambda_j x}, \qquad \lambda_1 < \cdots < \lambda_m, \tag{S147}$$

with no identically zero summand has at most $2m-1$ real zeros counted with multiplicity.

First, the zero set is finite. After multiplication by $\mathrm{e}^{-\lambda_1 x}$, the first affine term dominates as $x \to -\infty$; the last affine-exponential term dominates as $x \to +\infty$. Each dominating affine factor has a fixed sign sufficiently far in its tail. Thus all zeros lie in a compact interval. A nonzero real-analytic function has only finitely many zeros there.

Now induct on m. The $m=1$ case is the affine zero bound. For $m>1$, put $Q=\mathrm{e}^{-\lambda_1 x}P_m$. Two derivatives annihilate the first affine term. For every $j \geq 2$, the second derivative of $(a_j+b_jx)\mathrm{e}^{(\lambda_j-\lambda_1)x}$ is a nonzero affine polynomial times the same exponential, and the remaining positive exponents are distinct. If a C^2 function has $N \geq 2$ real zeros counted with multiplicity, generalized Rolle counting gives at least $N-2$ zeros of its second derivative counted with multiplicity: differentiation reduces the multiplicity at each root, and ordinary Rolle supplies a root between each pair of distinct roots. Hence

$$N(Q)-2 \leq N(Q'') \leq 2(m-1)-1, \tag{S148}$$

so $N(P_m) \leq 2m-1$; the cases $N<2$ are immediate. In Eq. (S146), every affine coefficient has nonzero slope $-w_j\mathrm{e}^{-c_j^2/(2\sigma^2)}$, so the claim applies. □

Lemma S5.4 (Uniform adjacent isolation)**.** *For the fixed finite centre set and the compact weight family, there are* $C_{\mathrm{iso}}, q_{\mathrm{iso}}>0$ *such that, for every* $j<m$*,*

$$\sup_{x\in[c_j,c_{j+1}]} \frac{\sum_{k\notin\{j,j+1\}} q_k(x)}{q_j(x)+q_{j+1}(x)} \leq C_{\mathrm{iso}}\mathrm{e}^{-q_{\mathrm{iso}}/\sigma^2} \tag{S149}$$

for all sufficiently small σ*, uniformly in* w*. On* $[\alpha, c_1]$*, component* 1 *has the analogous one-component bound, and on* $[c_m,\beta]$*, component* m *does.*

Proof. For $x \in [c_j, c_{j+1}]$ and $k<j$,

$$(x-c_k)^2-(x-c_j)^2=(c_j-c_k)(2x-c_j-c_k) \geq (c_j-c_k)^2. \tag{S150}$$

For $k>j+1$, comparison with c_{j+1} gives the corresponding positive lower bound. Since $w_k/w_j \leq 1/w_*$, summation over the finite centre set proves Eq. (S149). The same difference-of-squares comparison proves the two tail statements. The posterior mean and variance of the full mixture therefore differ from those of the isolated adjacent pair by $O(\mathrm{e}^{-q/\sigma^2})$, uniformly in the gap, the allocation, and sufficiently small σ. □

Theorem S5.5 (Exact topology of the pure mixture)**.** *Uniformly for* $w \in \mathcal{W}_{w_*}$*, every sufficiently small* $\sigma>0$ *gives exactly* m *simple local maxima and* $m-1$ *simple local minima of* $H_{\sigma,w}$ *on* J*, ordered alternately, with nonzero endpoint derivatives. The maximum near* c_j*, denoted* $p_j(\sigma,w)$*, satisfies*

$$p_j=c_j+O(\mathrm{e}^{-q/\sigma^2}), \qquad L'_{\sigma,w}(p_j)=-\sigma^{-2}+o(\sigma^{-2}). \tag{S151}$$

For $m \geq 2$*, define the weighted equality point*

$$s_j(\sigma,w)=\frac{c_j+c_{j+1}}{2}+\frac{\sigma^2}{\Delta_j}\log\frac{w_j}{w_{j+1}}. \tag{S152}$$

The minimum in the j*-th gap, denoted* $r_j(\sigma,w)$*, satisfies*

$$r_j=s_j+O(\sigma^4), \tag{S153}$$

$$L'_{\sigma,w}(r_j)=\frac{\Delta_j^2}{4\sigma^4}+O(\sigma^{-2}). \tag{S154}$$

All remainders are uniform on the compact weight set.

Proof. Fix a constant $A>1$. On $|x-c_j| \leq A\sigma^2$, one-component isolation gives

$$L_{\sigma,w}(x)=\frac{c_j-x}{\sigma^2}+O(\sigma^{-2}\mathrm{e}^{-q/\sigma^2}), \tag{S155}$$

$$L'_{\sigma,w}(x)=-\sigma^{-2}+O(\sigma^{-4}\mathrm{e}^{-q/\sigma^2}). \tag{S156}$$

At $c_j-\sigma^2$ the slope is positive, at $c_j+\sigma^2$ it is negative, and it is strictly decreasing between them. Hence there is one simple maximum with the location and derivative in Eq. (S151).

For a gap, the isolated adjacent odds are exactly

$$\frac{q_{j+1}(x)}{q_j(x)} = \exp\left[\frac{\Delta_j(x-s_j)}{\sigma^2}\right]. \tag{S157}$$

Put $h_j = (\log 9)/\Delta_j$ and

$$C_j = [s_j - h_j\sigma^2, s_j + h_j\sigma^2]. \tag{S158}$$

At the left and right endpoints the adjacent posterior masses are, respectively, $(9/10, 1/10)$ and $(1/10, 9/10)$. By Lemma S5.4, after reducing a common σ_0, both adjacent posterior masses are bounded below throughout C_j, and

$$\mathrm{Var}_{\pi(x)}(c) \geq \kappa_0\Delta_*^2, \qquad L'_{\sigma,w}(x) \geq \frac{\kappa_v}{\sigma^4} > 0 \quad (x \in C_j) \tag{S159}$$

for constants independent of j, w, σ. At the two endpoints of C_j, Eq. (S144) gives, for sufficiently small σ,

$$L_{\sigma,w}(s_j - h_j\sigma^2) \leq -\frac{\Delta_j}{5\sigma^2}, \qquad L_{\sigma,w}(s_j + h_j\sigma^2) \geq \frac{\Delta_j}{5\sigma^2}. \tag{S160}$$

Thus C_j contains exactly one simple minimum root.

At s_j, the adjacent posterior masses are equal and the adjacent posterior mean is $(c_j + c_{j+1})/2$. Hence

$$L_{\sigma,w}(s_j) = -\frac{1}{\Delta_j}\log\frac{w_j}{w_{j+1}} + O(\sigma^{-2}\mathrm{e}^{-q/\sigma^2}) = O(1) \tag{S161}$$

uniformly in w. The mean-value theorem and Eq. (S159) give $r_j - s_j = O(\sigma^4)$. At r_j, the two adjacent posterior masses are $1/2 + O(\sigma^2)$; substitution in Eq. (S145) proves Eq. (S154).

For small σ, the constructed peak and gap neighbourhoods are pairwise disjoint and contained in (α, β). They contain $m + (m-1) = 2m-1$ distinct simple stationary points. The global bound in Lemma S5.3 excludes every additional real stationary point, including multiple ones. Tail isolation and the fixed separations $c_1 - \alpha > 0$ and $\beta - c_m > 0$ give

$$L_{\sigma,w}(\alpha) > 0, \qquad L_{\sigma,w}(\beta) < 0, \tag{S162}$$

so neither endpoint is stationary. When $m = 1$, there are no gaps or valley definitions: the exact identity $L_{\sigma,w}(x) = (c_1 - x)/\sigma^2$ gives one maximum and both endpoint signs directly. □

S5.3. Full posterior-sector certificate and a slow positive factor

The zero bound above does not apply after multiplication by a nonconstant positive factor. We therefore certify the complete complement of small peak and valley boxes.

Lemma S5.6 (Posterior-sector certificate)**.** *Fix $K > 0$. There are constants $A_\mathrm{p}, A_\mathrm{v}, C_\mathrm{p}, C_\mathrm{v}, \kappa_\mathrm{p}, \kappa_\mathrm{v} > 0$ and $\sigma_K > 0$, depending only on the fixed centre set, J, w_*, and K, such that, for every $0 < \sigma < \sigma_K$ and every admissible w, the boxes*

$$P_j = [c_j - A_\mathrm{p}\sigma^2, c_j + A_\mathrm{p}\sigma^2], \tag{S163}$$

$$V_j = [r_j - A_\mathrm{v}\sigma^4, r_j + A_\mathrm{v}\sigma^4] \quad (j < m) \tag{S164}$$

are pairwise disjoint and contained in (α, β), and the following hold:

$$|L_{\sigma,w}| \leq C_\mathrm{p}, \qquad L'_{\sigma,w} \leq -\kappa_\mathrm{p}\sigma^{-2} \qquad \text{on } P_j, \tag{S165}$$

$$|L_{\sigma,w}| \leq C_\mathrm{v}, \qquad L'_{\sigma,w} \geq \kappa_\mathrm{v}\sigma^{-4} \qquad \text{on } V_j. \tag{S166}$$

At the left and right box boundaries, respectively, the slope signs are

$$\begin{array}{c|cc} & \textit{left boundary} & \textit{right boundary} \\ \hline P_j & L_{\sigma,w} \geq 2K & L_{\sigma,w} \leq -2K \\ V_j & L_{\sigma,w} \leq -2K & L_{\sigma,w} \geq 2K. \end{array} \tag{S167}$$

On the entire complement in J of the open boxes,

$$|L_{\sigma,w}(x)| \geq 2K, \tag{S168}$$

with the exhaustive alternating sign order

$$+\ P_1\ -\ V_1\ +\ P_2\ -\ \cdots\ +\ P_m\ -. \tag{S169}$$

For $m = 1$, every valley clause is omitted.

Proof. Choose $A_\mathrm{p} \geq 4K$. The uniform expansions (S155)– (S156) give, after reducing σ_K, Eq. (S165) and the peak row of Eq. (S167). The convex-hull property $\bar{c}(x) \in [c_1, c_m]$ gives the full outer-tail estimates without any local expansion:

$$\begin{aligned} x \leq c_1 - A_\mathrm{p}\sigma^2 &\Longrightarrow L_{\sigma,w}(x) \geq A_\mathrm{p}, \\ x \geq c_m + A_\mathrm{p}\sigma^2 &\Longrightarrow L_{\sigma,w}(x) \leq -A_\mathrm{p}. \end{aligned} \tag{S170}$$

Fix a gap and abbreviate $\Delta = \Delta_j$, $s = s_j$. The right adjacent posterior mass is

$$p(x) = \frac{q_{j+1}(x)}{q_j(x) + q_{j+1}(x)} = \frac{1}{1 + \exp[-\Delta(x - s)/\sigma^2]}. \tag{S171}$$

On the crossover interval C_j in Eq. (S158), its edge odds are $1/9$ and 9, not exponentially small or large. Both adjacent masses are therefore bounded below throughout C_j, and Eq. (S159) holds there. Conversely, the fixed finite centre diameter bounds the posterior variance, so $|L'| \leq C\sigma^{-4}$ on C_j.

The pure root satisfies $L(r_j) = 0$. Choose A_v large enough that $\kappa_v A_\mathrm{v} \geq 2K$. Since $A_\mathrm{v}\sigma^4 = o(\sigma^2)$, the valley box lies in C_j for small σ. Integrating the lower and upper bounds for L' from r_j proves Eq. (S166), the valley row of Eq. (S167), and the required negative and positive bounds on $C_j \setminus \operatorname{int} V_j$.

It remains to cover both outer sectors of every gap; this is the step that prevents an unproved dominance claim at a crossover edge. Consider first

$$[c_j + A_\mathrm{p}\sigma^2, s_j - h_j\sigma^2] \tag{S172}$$

and split it at $c_j + \Delta_j/4$, which lies in the stated interval for all sufficiently small σ, uniformly in w. On the portion nearer c_j, every other component is exponentially small relative to component j, so

$$\bar{c}(x) - x \leq -\tfrac{1}{2}A_\mathrm{p}\sigma^2, \qquad L_{\sigma,w}(x) \leq -A_\mathrm{p}/2 \leq -2K. \tag{S173}$$

On the remaining portion, monotonicity of Eq. (S171) and the left crossover edge give $p(x) \leq 1/10$, while $x - c_j \geq \Delta_j/4$. The adjacent-pair mean then obeys

$$c_j + \Delta_j p(x) - x \leq \frac{\Delta_j}{10} - \frac{\Delta_j}{4} = -\frac{3\Delta_j}{20}. \tag{S174}$$

The nonadjacent correction is exponentially small by Lemma S5.4; hence this sector also has $L \leq -2K$ for small σ, including its 1/9-odds edge.

Symmetrically, split $[s_j + h_j\sigma^2, c_{j+1} - A_\mathrm{p}\sigma^2]$ at $c_{j+1} - \Delta_j/4$. On the portion nearer the crossover, $p(x) \geq 9/10$ and $c_{j+1} - x \geq \Delta_j/4$, giving the positive analogue of Eq. (S174); on the portion nearer c_{j+1}, one-component isolation gives $L \geq A_\mathrm{p}/2 \geq 2K$. Thus the right outer sector has $L \geq 2K$.

The fixed centre and endpoint separations make all boxes disjoint and interior for one common sufficiently small σ_K. The outer tails, all peak boundaries, both outer sectors of every gap, and every crossover-minus-valley sector cover every point of the box complement. This proves Eqs. (S168)– (S169). When $m = 1$, the same result follows directly from $L = (c_1 - x)/\sigma^2$ and Eq. (S170). □

Theorem S5.7 (Slow positive factors preserve the exact topology)**.** *Let $x \in C^2(I)$ be strictly increasing with $\inf_I x' = v_0 > 0$. Let $a_\sigma \in C^2(I)$ be positive and assume*

$$M_0 := \sup_{\sigma<\sigma_0} \|\partial_t \log a_\sigma\|_{L^\infty(I)} < \infty, \qquad M_1 := \sup_{\sigma<\sigma_0} \|\partial_t^2 \log a_\sigma\|_{L^\infty(I)} < \infty. \tag{S175}$$

Then, uniformly for $w \in \mathcal{W}_{w_*}$, all sufficiently small $\sigma > 0$ give exactly m nondegenerate maxima and $m-1$ nondegenerate minima of

$$F_{\sigma,w}(t) = a_\sigma(t) H_{\sigma,w}(x(t)) \tag{S176}$$

on I, ordered alternately, with no stationary endpoint. Relative to the pure roots in the x coordinate,

$$x(t_{j,\sigma}^{\max}) - p_j(\sigma, w) = O(\sigma^2), \tag{S177}$$
$$x(t_{j,\sigma}^{\min}) - r_j(\sigma, w) = O(\sigma^4), \tag{S178}$$

uniformly in w. Thus

$$x(t_{j,\sigma}^{\min}) = s_j(\sigma, w) + O(\sigma^4). \tag{S179}$$

Proof. Put

$$b_\sigma = \partial_t \log a_\sigma, \qquad D_{\sigma,w}(t) = \partial_t \log F_{\sigma,w}(t) = b_\sigma(t) + x'(t) L_{\sigma,w}(x(t)). \tag{S180}$$

Since $F_{\sigma,w} > 0$, its stationary points are precisely the zeros of $D_{\sigma,w}$, and

$$D'_{\sigma,w} = b'_\sigma + x'' L_{\sigma,w} + (x')^2 L'_{\sigma,w}. \tag{S181}$$

Choose $K > (M_0 + 1)/v_0$ and apply Lemma S5.6. On the full complement of the open boxes, the term $x'L$ strictly dominates b_σ, so D has the alternating signs in Eq. (S169) and has no complement zero.

On a peak box, Eqs. (S165) and (S181) give

$$D'_{\sigma,w} \le M_1 + \|x''\|_\infty C_{\mathrm{p}} - v_0^2 \kappa_{\mathrm{p}} \sigma^{-2} < 0 \tag{S182}$$

for small σ. Its boundary signs give exactly one zero. At that zero $F'' = FD' < 0$, hence the root is a nondegenerate maximum. Similarly, on a valley box,

$$D'_{\sigma,w} \ge -M_1 - \|x''\|_\infty C_{\mathrm{v}} + v_0^2 \kappa_{\mathrm{v}} \sigma^{-4} > 0, \tag{S183}$$

so its unique zero is a nondegenerate minimum. The certified tail signs exclude endpoint stationary points.

At a slow-factor root, Eq. (S180) gives $|L| \le M_0/v_0$. The pure peak root has $L(p_j) = 0$, and the uniform negative $L' = \Theta(\sigma^{-2})$ on its box implies the peak displacement in Eq. (S177). The uniform positive $L' = \Theta(\sigma^{-4})$ on a valley box gives Eq. (S178). Combining the latter with Eq. (S153) proves Eq. (S179). Every constant is uniform over the compact allocation family. □

S5.4. Fixed-ε Doi transfer and theorem boundary

For $B > 0$, let

$$f_{B,\varepsilon,w}(t) = \mathbb{E}_0\left[K_{B,w,\varepsilon}(Z_t, R_t) \exp\left(-\int_0^t K_{B,w,\varepsilon}(Z_s, R_s)\,\mathrm{d}s\right)\right] \tag{S184}$$

be the Doi reaction-time density and set

$$\mathcal{F}_{B,\varepsilon,w} = f_{B,\varepsilon,w}/B. \tag{S185}$$

This is budget-rescaling, not normalization to unit time integral.

Theorem S5.8 (Exact m Doi modes for fixed finite (d, m))**.** *Fix all data in Eqs.* (S116)– (S136)*, including the stationary midpoint variance, mutual independence, the strict weighted-space variance inequalities, target times strictly inside I, consistently oriented physical centres, normalized slabs of physical width $\varepsilon\rho$, the whole-window contact margin, and the compact allocation family. Then there is $\varepsilon_0 > 0$, depending on these fixed data, such that for every $0 < \varepsilon < \varepsilon_0$ and every $w \in \mathcal{W}_{w_*}$, the continuum free-exposure clock $G_{\varepsilon,w}$ has exactly m nondegenerate maxima and $m-1$ nondegenerate minima on I, ordered alternately, and no stationary endpoint.*

For each fixed such ε, there is $B_0(\varepsilon) > 0$, uniform over $w \in \mathcal{W}_{w_*}$, such that $\mathcal{F}_{B,\varepsilon,w}$ has the same complete stationary signature for every $0 < B < B_0(\varepsilon)$. Multiplication by $B > 0$ changes no stationary point, so the statement also holds for the physical density $f_{B,\varepsilon,w}$.

Proof. In Lemma S5.2, take $a_\sigma = a_\varepsilon$, with $\sigma = \varepsilon S_*/\ell_0$. Lemma S5.1 supplies Eq. (S175); hence Theorem S5.7 proves the exact stationary signature of $G_{\varepsilon,w}$, uniformly over the compact allocation family.

Fix one $0 < \varepsilon < \varepsilon_0$. The ordered simple roots depend continuously on w by the implicit-function theorem. Their finitely many graphs over the compact set $\mathcal{W}_{w_*}$ are compact and disjoint. Continuity of the second time derivative gives pairwise disjoint root tubes on which G'' has a uniform nonzero sign and magnitude. After shrinking the tubes if necessary, G' has fixed opposite signs at the two boundaries of each tube. On the compact complement in $I \times \mathcal{W}_{w_*}$, $|G'|$ has a positive minimum; the two endpoint slopes have positive uniform absolute minima as well.

For this fixed ε, $V_{w,\varepsilon}$ is bounded uniformly in w, and Eqs. (S119)– (S121) put q_0 in the required weighted state space. Because $w_j \geq w_* > 0$, the compact allocation set lies strictly inside the simplex and admits the affine control chart and a complex tube required by Theorem S3.4. (If the set is the singleton $w_j = 1/m$, the same conclusion is the corresponding pointwise case.) Applying that theorem with time orders $0, 1, 2$ and no control derivative gives

$$\sup_{w \in \mathcal{W}_{w_*}} \|\mathcal{F}_{B,\varepsilon,w} - G_{\varepsilon,w}\|_{C^2(I)} \longrightarrow 0 \qquad (B \downarrow 0). \tag{S186}$$

Choose $B_0(\varepsilon)$ so that the C^1 error is smaller than all boundary, complement, and endpoint margins and the second-derivative error is smaller than the root-tube curvature margin. In each tube, $\mathcal{F}'$ has opposite boundary signs and $\mathcal{F}''$ keeps the sign of G'', so it has exactly one root of the same type. The complement and endpoints contain none. This proves the complete finite-window signature. □

Remark S5.9 (Quantifiers, saturation, and scope). *The quantifier order is*

$$\textit{fix } (d, m, \textit{all geometric and probabilistic data}) \implies 0 < \varepsilon < \varepsilon_0 \implies 0 < B < B_0(\varepsilon). \tag{S187}$$

There is no asserted lower bound for $B_0(\varepsilon)$ uniform as $\varepsilon \downarrow 0$, and no interchange of the two limits. The construction is not uniform in d or m, does not provide one geometry with arbitrarily many modes, does not count stationary points outside I, and gives no process-level event-mass floor. It treats normalized longitudinal Gaussian slabs, not arbitrary localized catalyst patches.

Moreover, Lemma S5.1 shows that the relative contact factor is asymptotically saturated on the theorem window:

$$c_{d,\varepsilon} = 1 + O(\varepsilon^{-N} \mathrm{e}^{-q/\varepsilon^2}) \quad \textit{in } C^2(I). \tag{S188}$$

Thus the exact Doi embedding is rigorous, but the modal basis is created by the ordered narrow slabs sampled by the monotone constant-variance midpoint; the relative encounter factor is a slow perturbation in this asymptotic family. Nontrivial contact, a useful common positive budget, deterministic finite-parameter complement certificates, survival, and event-basin masses remain separate continuum-validation requirements rather than consequences of Theorem S5.8.

S6. CONTROL-FREE FINITE-VOLUME SOURCE AND FREE-AXIS BOUNDARY

The physical two-dimensional finite-volume program fixes, independently of the control and reaction budget, the three-coordinate quotient initial law

$$\begin{aligned} b(u) &= \mathbf{1}_{\{|u|<1\}} \exp[-(1-u^2)^{-1}], & I_b &= \int_{-1}^{1} b(u)\,\mathrm{d}u, \\ \beta_{h,c}(x) &= \frac{b((x-c)/h)}{hI_b}, & q_0(M,R,Y) &= \beta_{h,M_0}(M)\beta_{h,R_0}(R)\beta^{\mathrm{per}}_{h,Y_0}(Y), \end{aligned} \tag{S189}$$

Here the periodic factor is defined on $0 \leq y < W$ by the locally finite image sum

$$\beta^{\mathrm{per}}_{h,c}(y) = \sum_{n \in \mathbb{Z}} \beta_{h,c}(y + nW). \tag{S190}$$

The source parameters use the following binary64 words, interpreted as exact dyadic reals:

$$\begin{aligned} M_0 &= \texttt{0x1.1eb851eb851ecp-3}, & R_0 &= \texttt{-0x1.6666666666666p-2}, \\ Y_0 &= 0, & h &= \texttt{0x1.47ae147ae147bp-6}, \qquad W = 1. \end{aligned} \tag{S191}$$

The canonical control-free source SHA-256 is obtained by concatenating the following two lines in order:

$$\begin{gathered} \texttt{0b2efec5dc1abea1380ab862e46825e7b} \\ \texttt{79658fe9bfa0ac6637e1426ed9f7f5f}. \end{gathered} \tag{S192}$$

TABLE I. Frozen control-free production configurations. The last column is the tensor state count; the sum is 34,787,462.

Row	N_M	N_R	N_Y	States
O113/Base	113	113	113	1,442,897
E128/Base	128	128	128	2,097,152
O129/Base	129	129	129	2,146,689
O161/Base	161	161	161	4,173,281
M+	166	129	129	2,762,406
R+	129	172	129	2,862,252
MR+	166	172	129	3,683,208
MR+F	207	215	161	7,165,305
A_M	129	128	128	2,113,536
A_R	128	129	128	2,113,536
A_Y	128	128	128	2,097,152
A_{MRY}	129	129	128	2,130,048

The nonperiodic domains use the exact binary64 endpoints

$$\begin{aligned}[L_M, U_M] &= [\texttt{-0x1.0000000000000p-2}, \texttt{0x1.d99999999999ap+0}],\\ [L_R, U_R] &= [\texttt{-0x1.ccccccccccccdp+0}, \texttt{0x1.ccccccccccccdp+0}],\end{aligned} \tag{S193}$$

and the periodic domain is $[L_Y, U_Y) = [0, 1)$. With $\Delta_a = (U_a - L_a)/4$, cell j on axis a is $[L_a + j\Delta_a, L_a + (j+1)\Delta_a)$, except that the last nonperiodic cell includes its right endpoint. For indices $j_M, j_R, j_Y \in \{0, 1, 2, 3\}$, the tensor component has C-order flat index

$$i = 16j_M + 4j_R + j_Y. \tag{S194}$$

The verifier reconstructs this exact four-cell partition on each axis, uses one shared directed-arithmetic enclosure of I_b, encloses every cell mass by composite Simpson quadrature with 16384 panels per unit and 192-bit directed exponentials, and forms the tensor endpoints in C order. It then rebuilds the packed bytes and the lower-anchor target from the source; a different but internally valid unit-mass box is rejected by byte identity. The resulting exact lower-anchor radius is

$$1 - \sum_i \ell_i = \frac{6051}{2^{53}} = 6.717959522006\ldots \times 10^{-13} < 10^{-12}. \tag{S195}$$

For this deliberately chosen verification partition, source containment also has a quadrature-independent proof. The midpoint support lies wholly in cell 0, the parallel-relative support lies wholly in cell 1, and evenness of b splits the periodic bump exactly across the quotient cut. Hence the three marginal vectors are

$$(1, 0, 0, 0), \qquad (0, 1, 0, 0), \qquad (1/2, 0, 0, 1/2), \tag{S196}$$

so the exact product target has C-order flat components 4 and 7 equal to 1/2, with all other components zero. An independently implemented directed-MPFR monotone-rectangle enclosure, using 8192 panels per unit and 224-bit arithmetic, checks all marginal and tensor components for numerical consistency; its much wider overlap is not used to prove canonical endpoint identity.

The preceding 4^3 construction is only the exact structural preflight. The same accepted source has now been applied, without any control or reaction budget, to the complete prospectively fixed production family in Table I. The configuration record fixes the axis alignment, exact domain or periodic shift, size, and row order before any positive-budget value is read.

For every row, the producer reconstructs the exact finite-volume partition, including vertex-centred endpoint half volumes and the prescribed half-period shift. It writes canonical big-endian binary64 endpoint files for the forward rate, backward rate, the historical `stationary_mass` role, and each initial marginal. The `stationary_mass` entries are the ungauged representative quadrature primitives

$$\widetilde{\mu}_{a,i} = |C_{a,i}| \exp[-\Phi_a(x^{\mathrm{rep}}_{a,i})],$$

not physical cell integrals and not box-normalized probabilities. Only active tensor components are stored; every omitted component is the exact positive-zero interval. The 206-file inventory occupies approximately 1.44 MB while its virtual dense endpoint stream has 556,599,392 bytes. The verifier rederives every partition, directed rate, primitive, marginal, sparse record, virtual dense digest, and row/family relation.

A same-core artifact implementation reconstructs all 206 files byte for byte. A third, separate-source implementation uses 256-bit directed MPFR arithmetic, an independently written Scharfetter–Gummel enclosure, and 32768 Simpson panels per unit. Its intervals lie inside the producer envelopes for all 10,074 directed rates, 5,037 ungauged representative primitives, 5,037 initial marginal components, and 722 active tensor components. Because both paths use the same `gmpy2`/MPFR backend and the same Simpson remainder lemma, this is separate-source semantic containment, not backend independence or an alternative quadrature family.

Separate authenticated fixed-row artifacts now compute the physical axis-cell integrals and reconstruct the formula-defined common edge fluxes, directed rates, and factorized density ratios. Their builders and independently written validators are executed from authenticated source snapshots under a pinned MPFR runtime. They still use the same backend, apply only to the twelve fixed level-$n = 0$ rows, and do not issue one correlated receipt containing a common mass, flux, gauge, map, and killing member.

The canonical forward/backward files are decoded and repacked into the native immutable rate format without arithmetic. All 72 conversions are round-tripped to the original big-endian bytes and retain the exact partition and axis-relation hashes. This binds packed free-axis rate payloads; it does not bind contact killing or a full generator.

An outer standard-library orchestrator runs two complete initial-bundle repeats, each using five isolated Python processes. A further two-repeat outer replay authenticates only the control-free factorized killing geometry. It constructs no control row, positive budget, tensor killing diagonal, reconstructed multiplier, or globally gauged operator.

Finally, a separate ideal authority defines twelve genuine dyadic refinement sequences with $h_f(n) = h_f(0)2^{-n}$. Its level-$n = 0$ equality with the saved configurations is geometric only. A source-bound ideal theorem candidate supplies symbolic map, contact cut-layer, and killing-residual bounds on those tails. It does not numerically evaluate the constants or supply production same-member containment.

There is therefore still no concrete killed `PackedKernelInputs` object, full operator, uniformization step, propagated target, stationary-point exclusion, measured production resource gate, F0 acceptance, project-level complete C1/C2, continuum transfer, or publication-level physical result.

S7. FORMAL-VERIFICATION BOUNDARY

The proofs in Secs. S3, S4, and S5 are conventional human-audited mathematical proofs. They are not Lean-verified. At this repository snapshot the companion modules reside under the exact repository-relative directory `research/reports/ring_lazy_jump_ext_rev2/code/formal_lean/FormalLean/`. The three source files and their SHA-256 anchors are listed below:

`Encounter.lean` : `d2c11759c831228eb6641f3944d1d860c34615982d15b883e6d029f0a670e754`,

`EncounterDesign.lean` : `fa45ceb3c40e7c9769d4f7d6ab5aa1495e89a361c675b89f362dfc11798b8330`,

`EncounterContinuum.lean` : `ae23060be3166c392eab2d8a0a5af5dcd1d3a4adf2a8b912fd8a0c2161e538b4`.

These files certify finite algebraic fragments only. Their declared scope excludes the continuum semigroup, differentiated wrapped-Gaussian tail, interval root count, finite-width/discretization persistence, and implicit-function/catastrophe bridge used here. In particular, a successful axiom report for those modules does not imply either Theorem S3.4 or Theorem S4.4.

A future formalization could first encode the tractable last mile: strict endpoint slopes plus strict concavity imply one unique nondegenerate maximum; uniform C^2 perturbations preserve those inequalities; and the nested quantifier wrapper fixes ϵ before B. Full formal verification would additionally require imported or formal proofs of the analytic-semigroup and differentiated wrapped-Gaussian estimates. Until then, the phrases “Lean verified” and “formally verified” are not admissible descriptions of the analytical theorems.

S8. RESERVED SUPPLEMENTAL MODULES

a. Reduced-clock/GIG ancestry placeholder. No GIG result or GIG-to-Doi universality enters Theorem S4.4. Any retained reduced-clock ancestry needs a separately audited, overlap-cleared account; this placeholder makes no claim.

TABLE II. Analytical evidence boundary for this working supplement.

Statement	Paper-proof status	Lean status
Affine budget projection and selected finite derivative identities	proved	partially covered by companion algebra
Analytic killed semigroup and mixed-jet $O(B)$ bridge	proved under stated hypotheses	not formalized
Differentiated wrapped-Gaussian contact tail	proved under stated hypotheses	not formalized
Local at-least-m, ϵ-then-B theorem in Sec. S4	proved under stated hypotheses	not formalized
Exact-m complete topology and fixed-ϵ Doi transfer in Sec. S5	complete reader-facing proof; frozen mathematical migration passed Round 149	not formalized
Fixed-row preparation, physical-integral/raw-flux sources, and ideal refinement/source-bound records	verified only at their respective control-free or ideal scopes; no correlated production member, propagation, complete C1/C2, or F0	not formalized
Finite-parameter cusp, event-mass floor, and solver convergence	not claimed in this supplement	no coverage claimed

b. Numerical-evidence placeholder. Apart from the control-free source/partition/free-axis preparation in Sec. S6, this file contains no numerical value or claim about positive-budget topology, cusps, mesh convergence, event mass, or independent physical solvers. That preparation is a method and provenance result, not a physical reaction-time result. Any positive-budget numerical module requires a frozen, independently checked discovery/confirmation/held-out chain; the analytical statements here rest only on their displayed hypotheses and proofs.

c. Exact-m proof status. Section S5 now contains the reader-facing migration of the accepted global zero bound, posterior-sector complement certificate, slow-factor root shifts, and fixed-ϵ weak-budget transfer. The frozen mathematical migration passed the independent hash-specific Round 149 audit. Submission remains barred because the finite-parameter F0–F3 evidence, convergence from the finite-volume scheme to the unbounded continuum (C0–C3 plus root transfer), overlap disclosure, metadata, and final release audits remain open; Section S4 remains only at its distinct, weaker at-least-m scope.

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